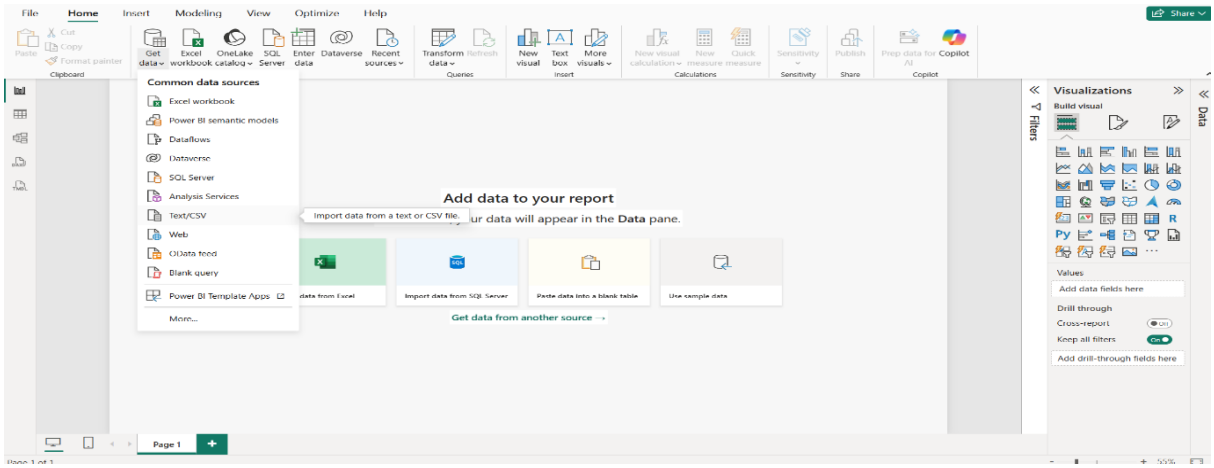
Data Transformation & Data Modeling

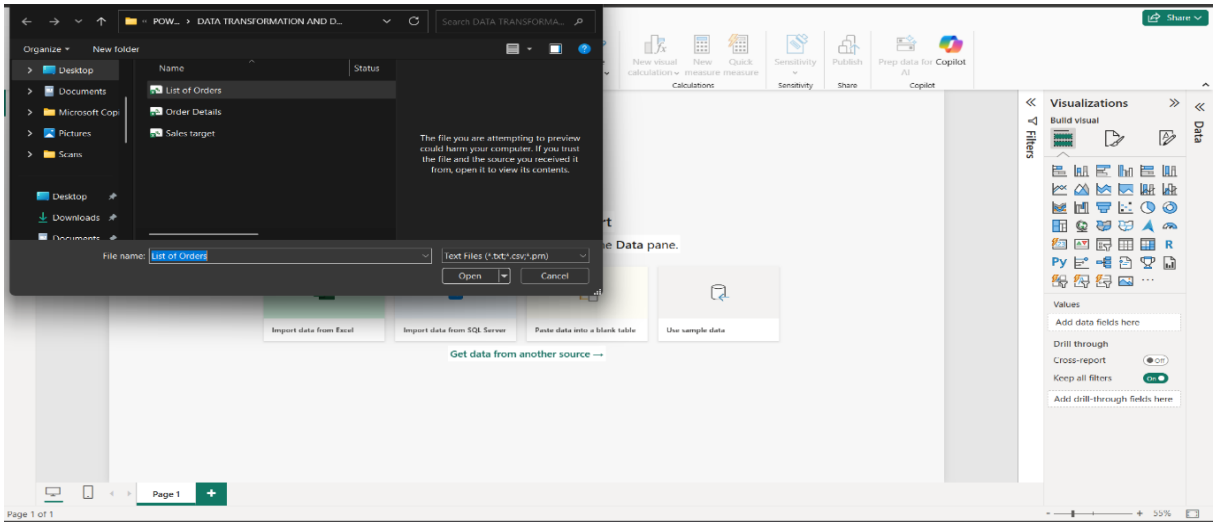
In this Assignment 3 CSV (Comma Seperated Value) E-Commerce Analysis files are given as a Dataset named .,

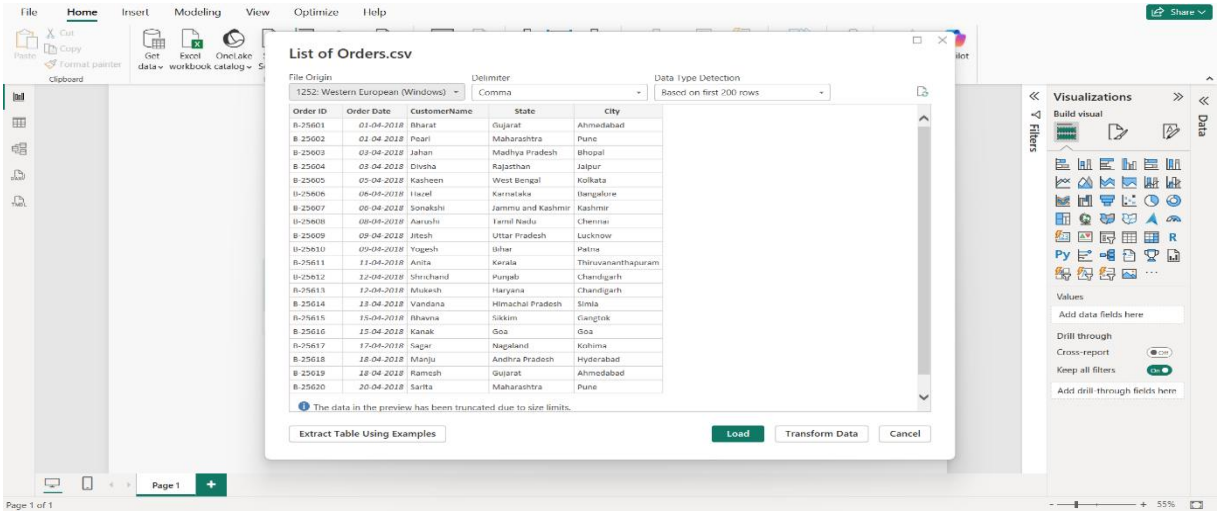
- List of Orders.CSV
- Order Details.CSV
- Sales Target.CSV

STEP-1----------IMPORTING DATA----------

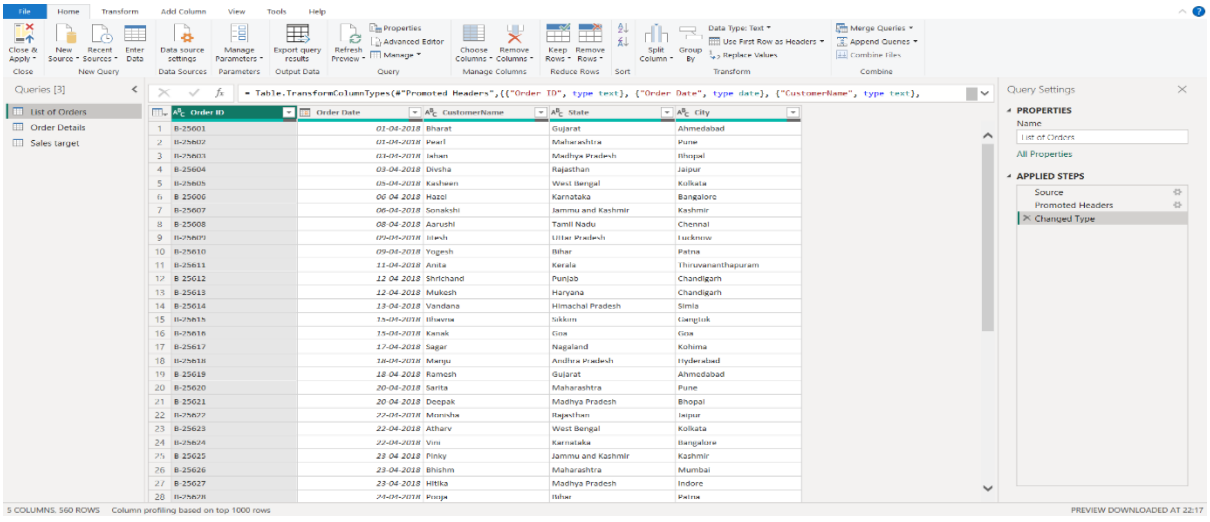
The given 3 CSV files are imported into POWERBI and then into the POWER QUERY EDITOR.







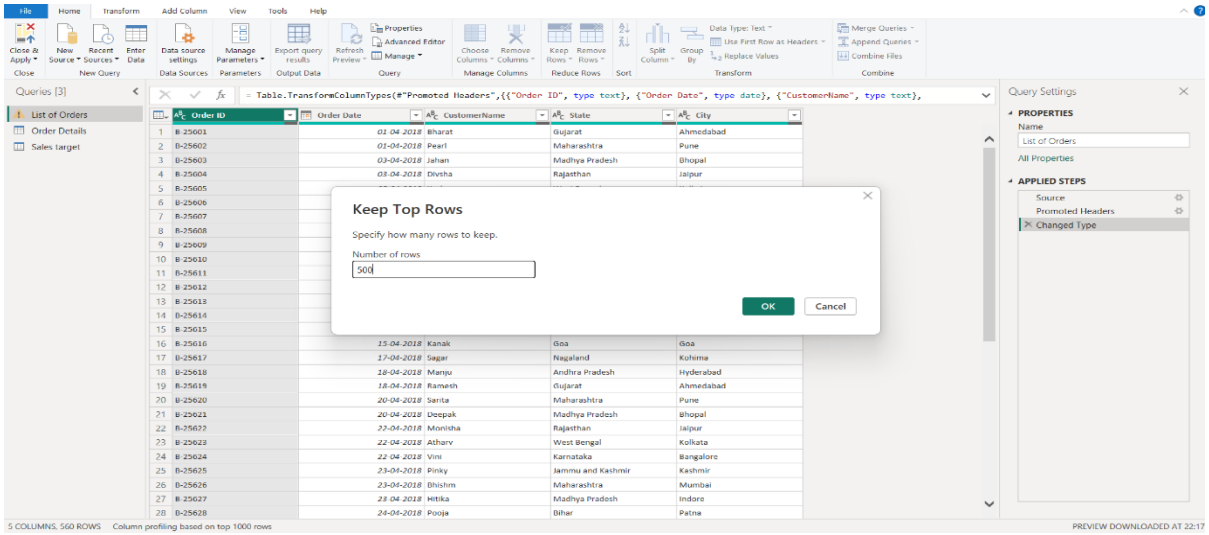
After the LOAD option is clicked , the LIST OF ORDERS.CSV file is loaded into the POWER QUERY EDITOR. The same procedure is repeated for other 2 files and are imported into the POWER QUERY EDITOR.

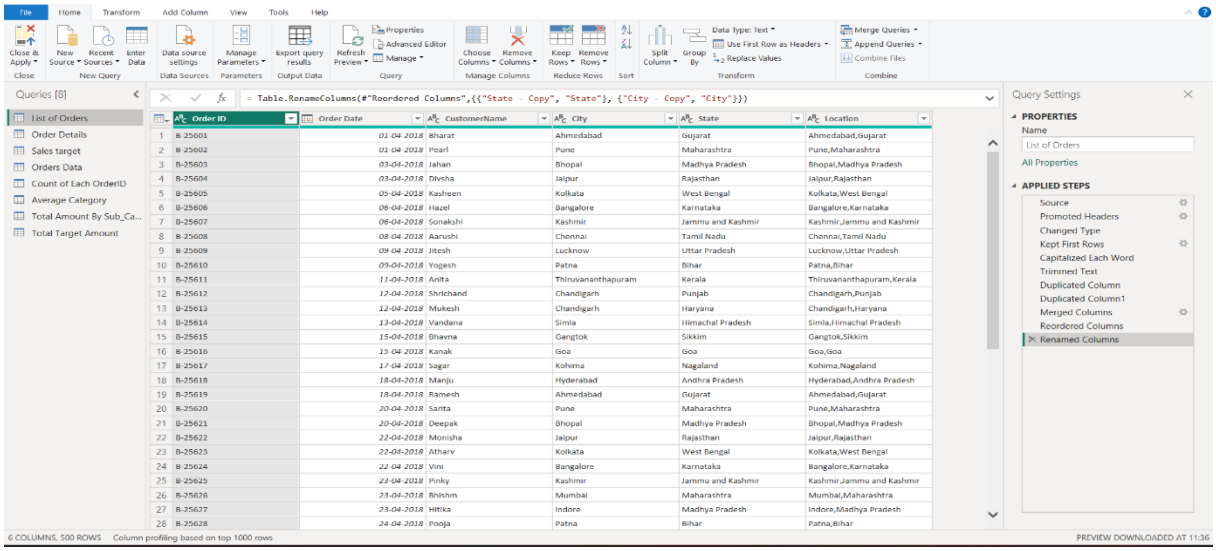


The given 3 CSV files are imported into POWERBI and POWER QUERY EDITOR.

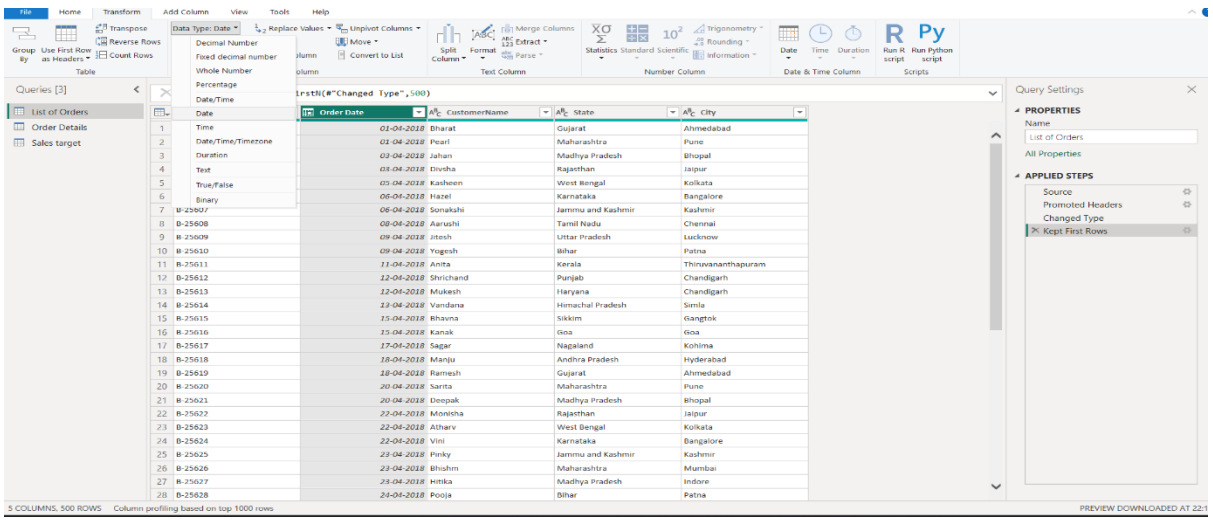
STEP-2----------DATA TRANSFORMATION----------

A) The LIST OF ORDERS table is restricted to the first 500 rows.

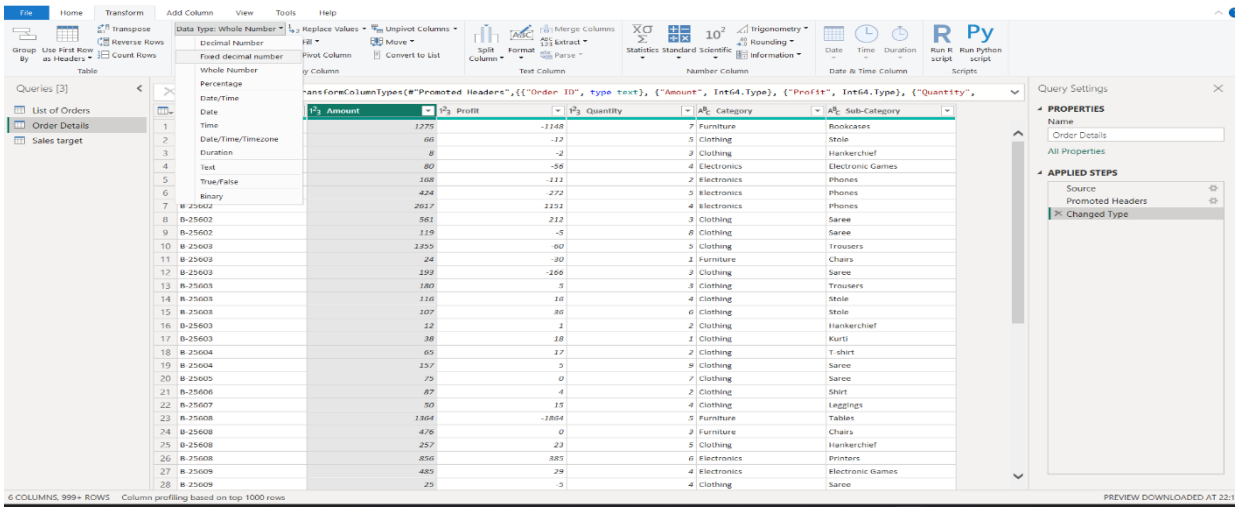


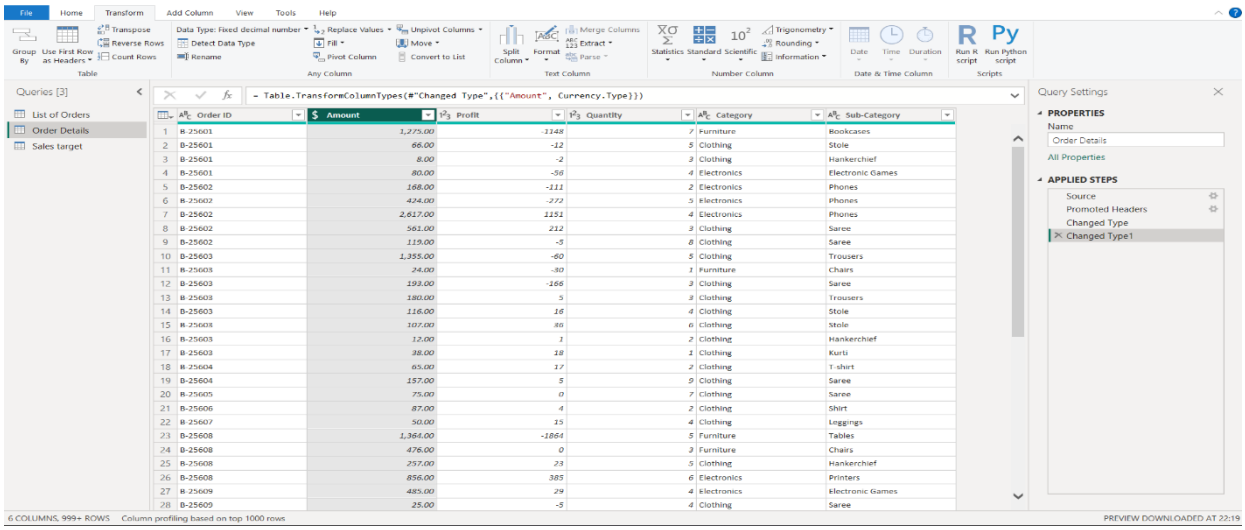


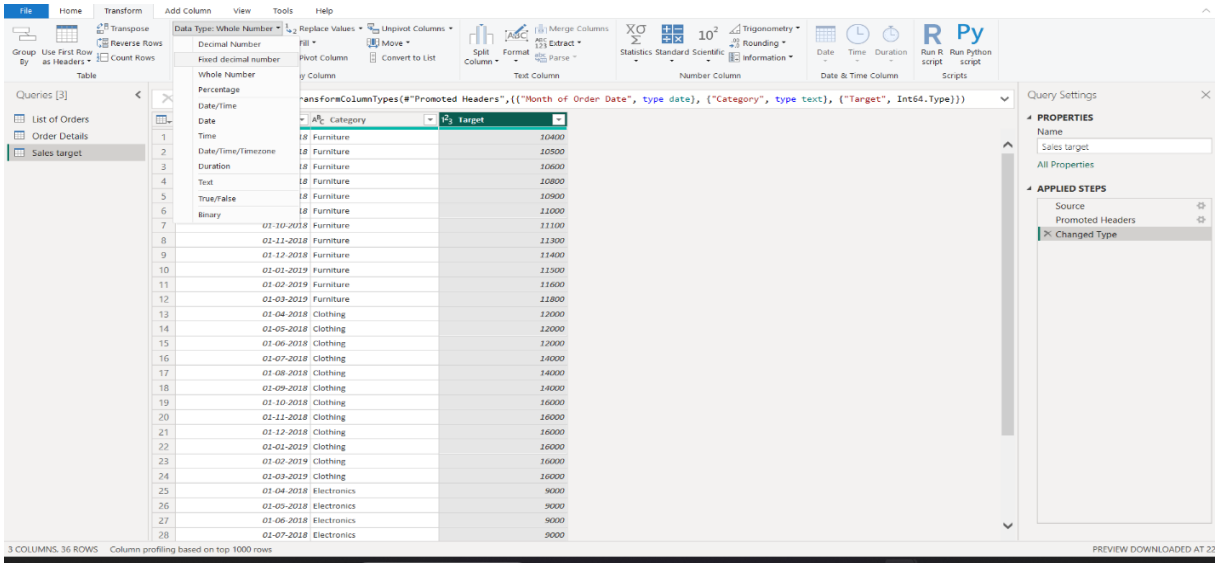
B) The datatype of the column ORDER DATE is ensured to be as DATE .
ORDER DATE-----DATE

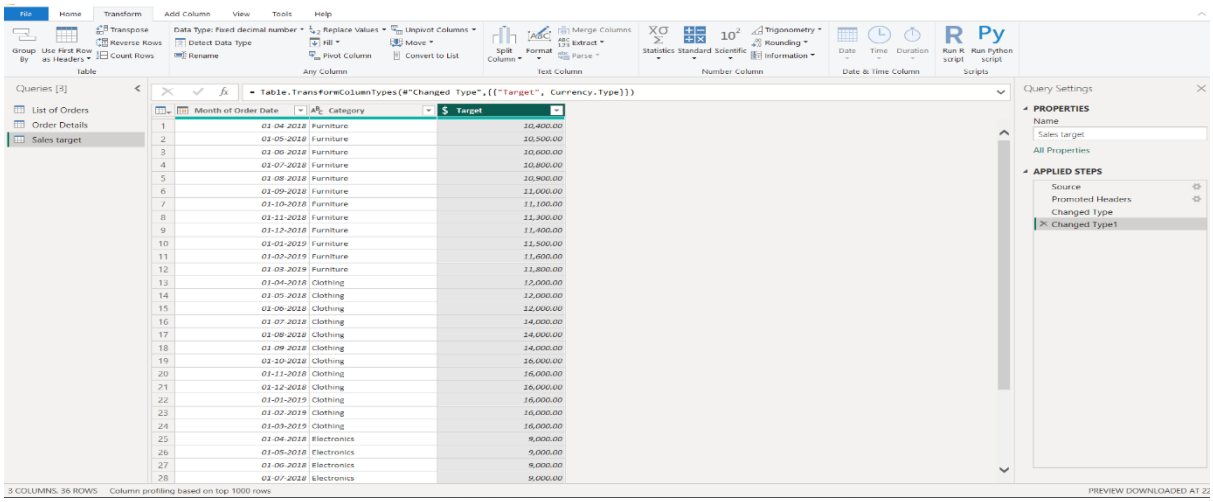


C) The datatype of the two columns are changed to,.
AMOUNT-----FIXED DECIMAL NUMBER
TARGET-----FIXED DECIMAL NUMBER

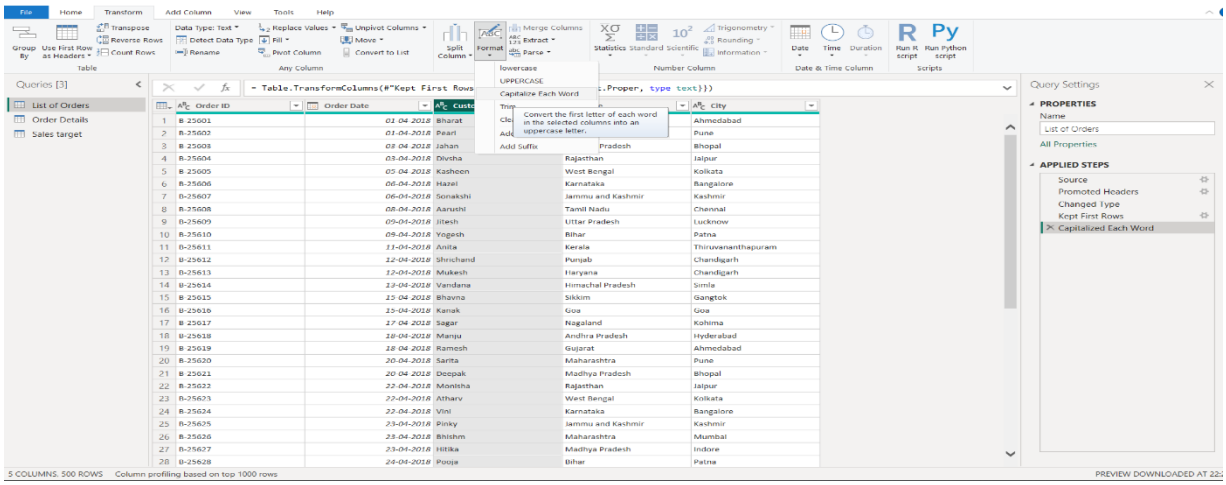




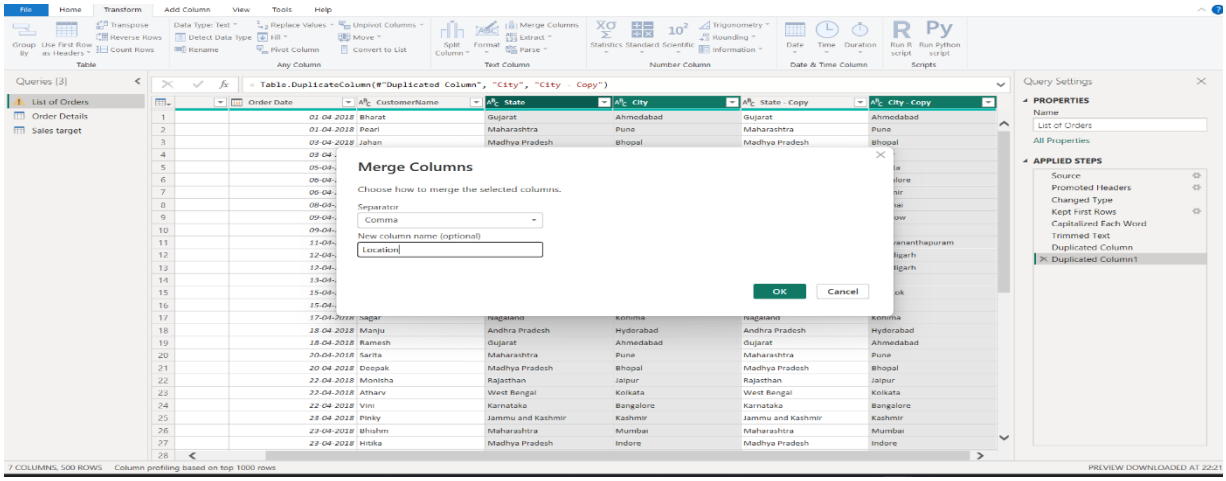


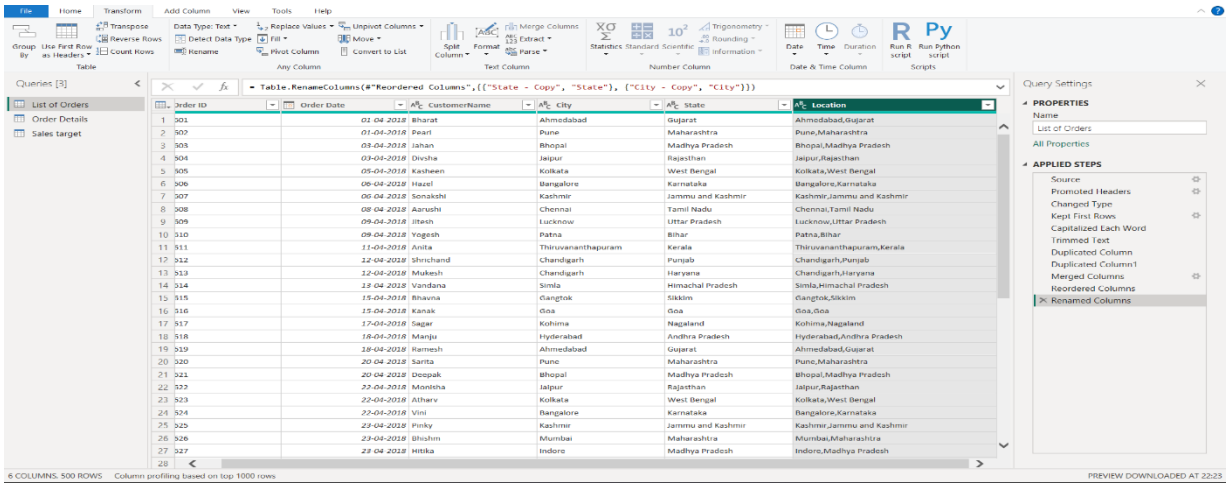


D) The column CUSTOMERNAME is Formatted into PROPER case.

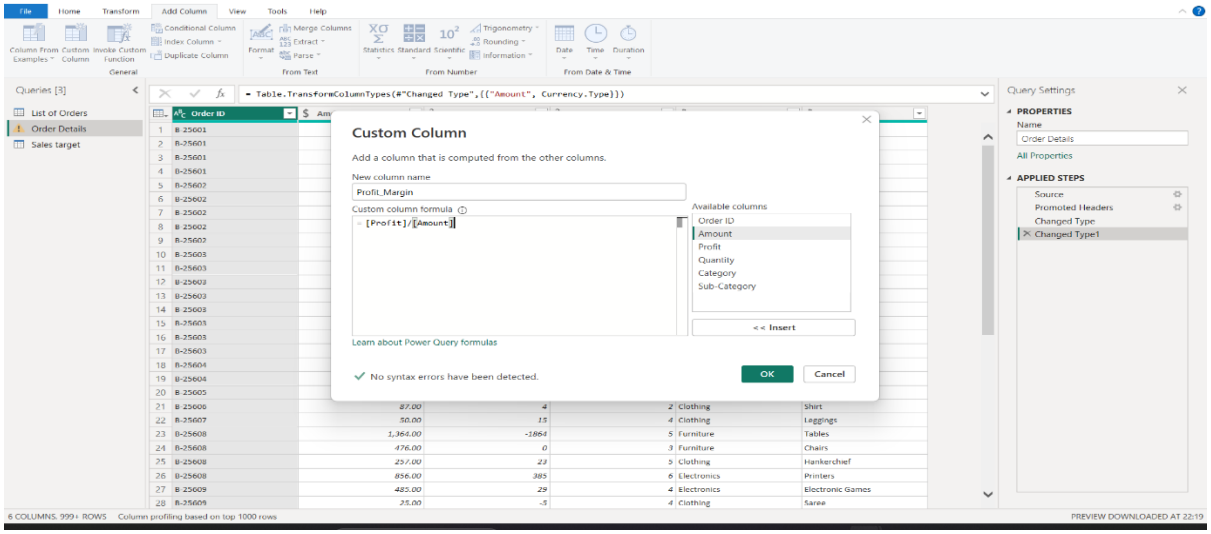


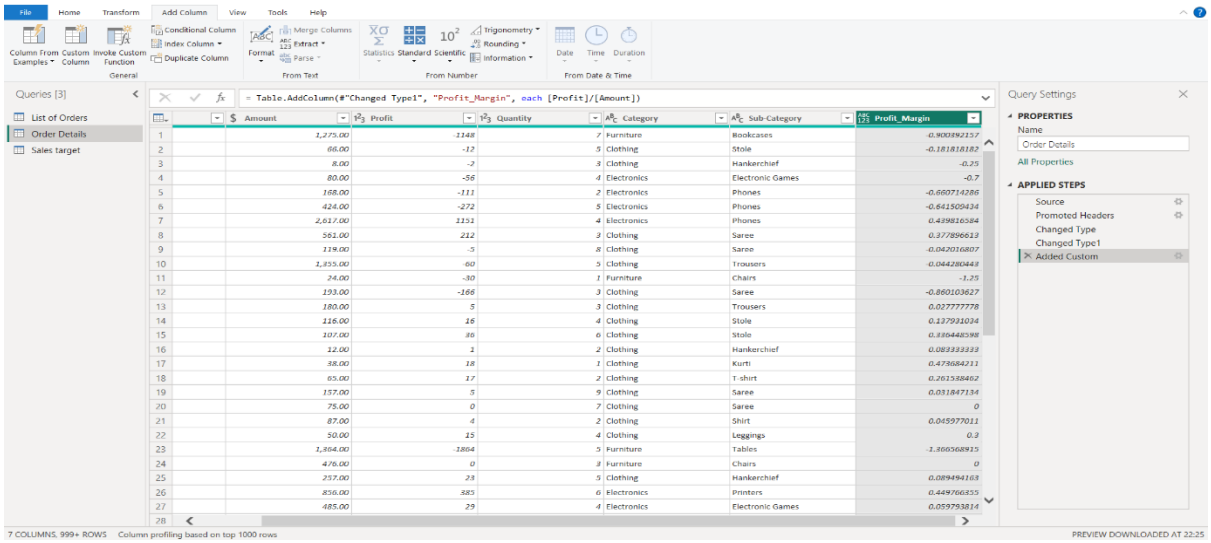
E) The columns STATE and CITY are merged in the format 'CITY , STATE' and the new column is named as LOCATION .





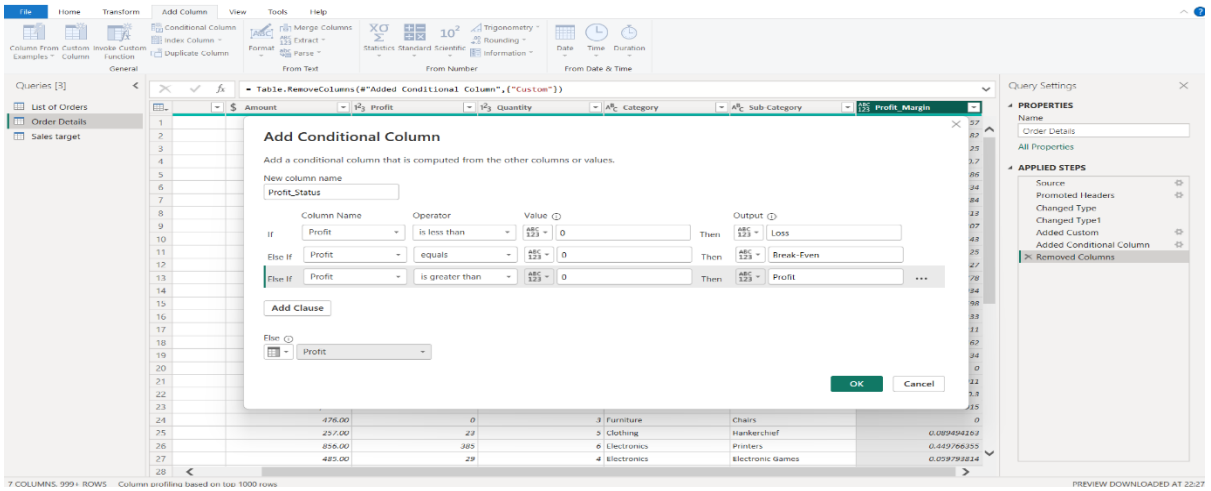
F) The new custom column named PROFIT_MARGIN is created as the result of the expression PROFIT / AMOUNT.

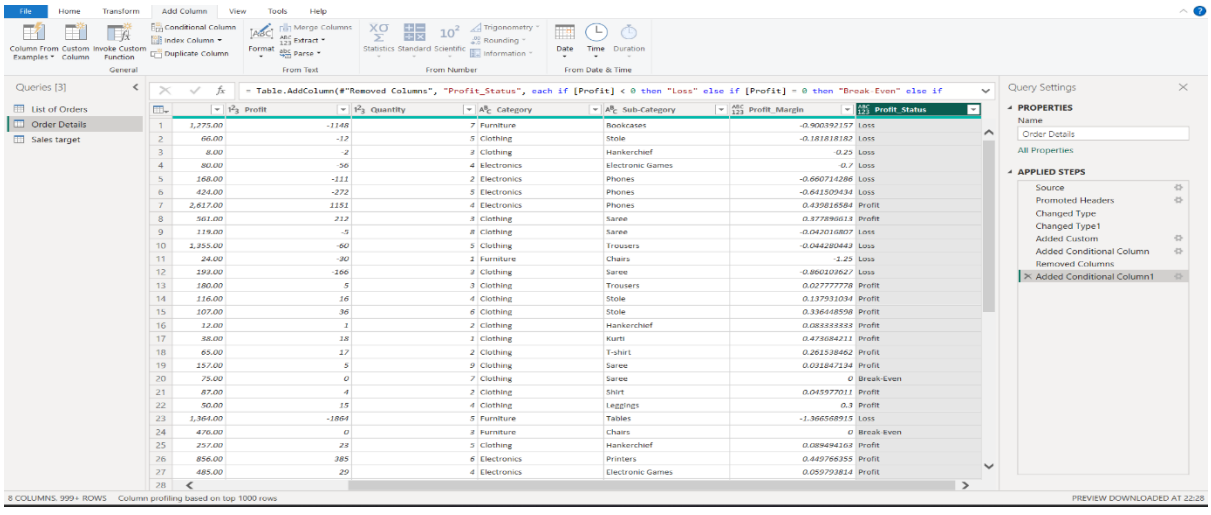




G) The new conditional column is created and named as PROFIT_STATUS with the following condition :

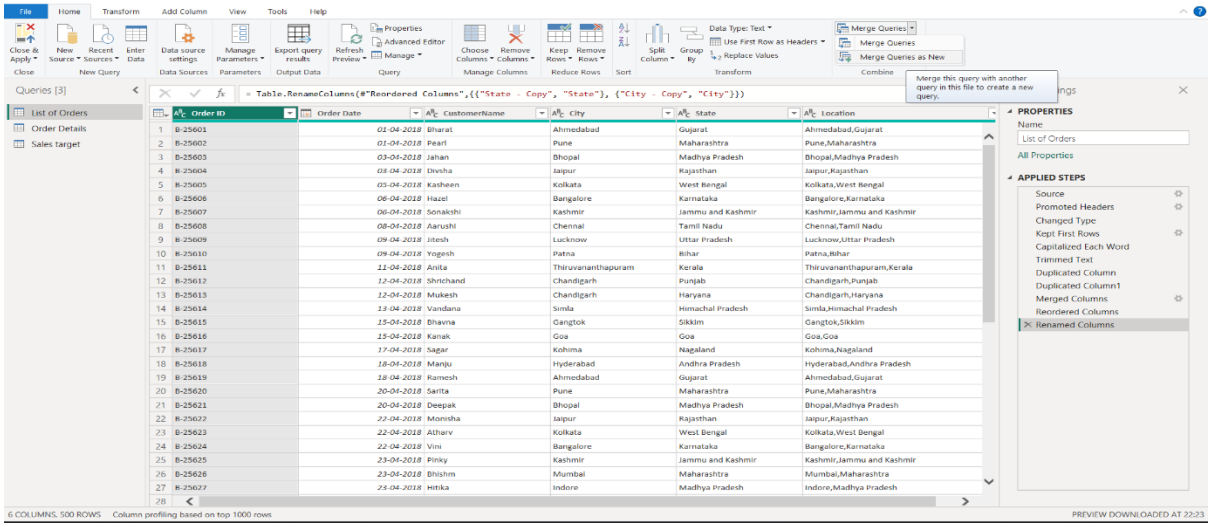
If the profit is less than 0, the label should be "Loss"; if the profit equals 0, the label should be "Break-Even"; and if the profit is greater than 0, the label should be "Profit".

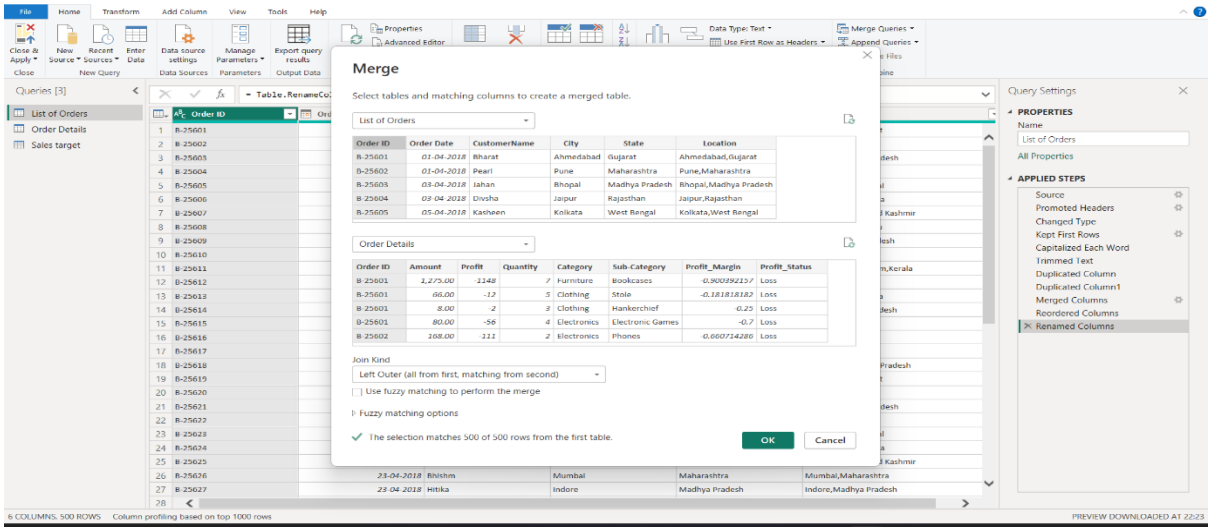


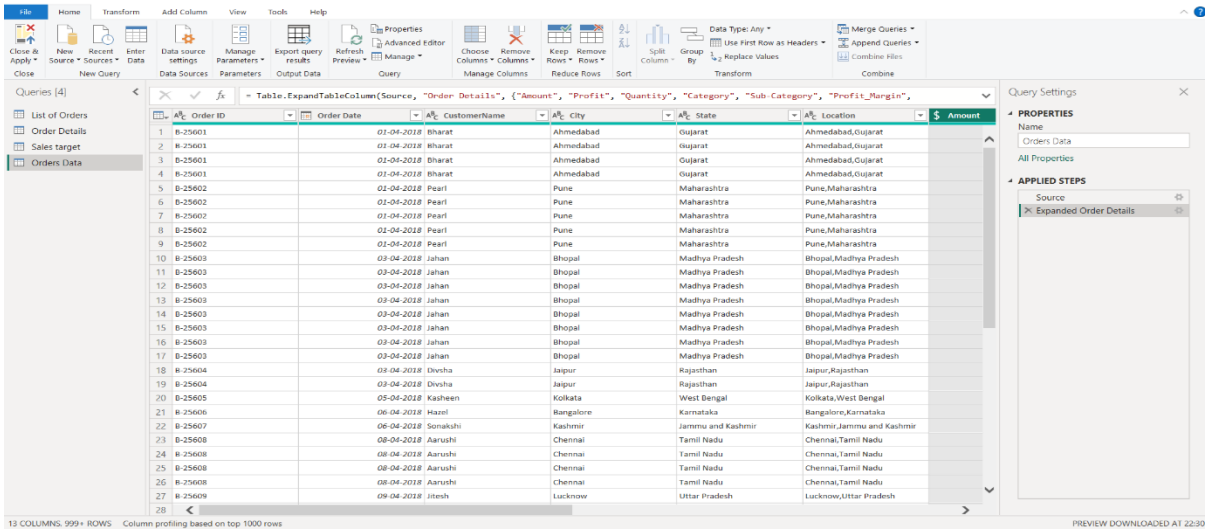


STEP-3----------MERGING DATA (JOINS)----------

The Tables LIST OF ORDERS and ORDER DETAILS are merged into a new single table and named as ORDERS DATA based on the column ORDER ID relationship.
In the Power query Editor, Go to the HOME tab , in the COMBINE group, select Merge Queries---Merge Queries as New.
The tables are selected and the ORDERID column in both the tables are selected and the LEFT OUTER JOIN is applied here.







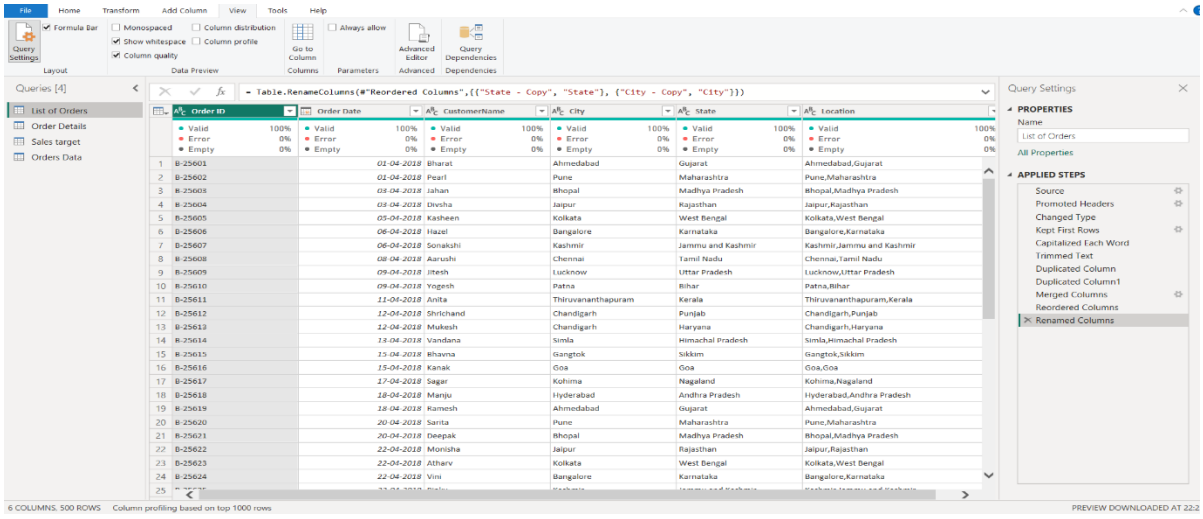
STEP-4----------HANDLING MISSING DATA & DUPLICATE DATA----------

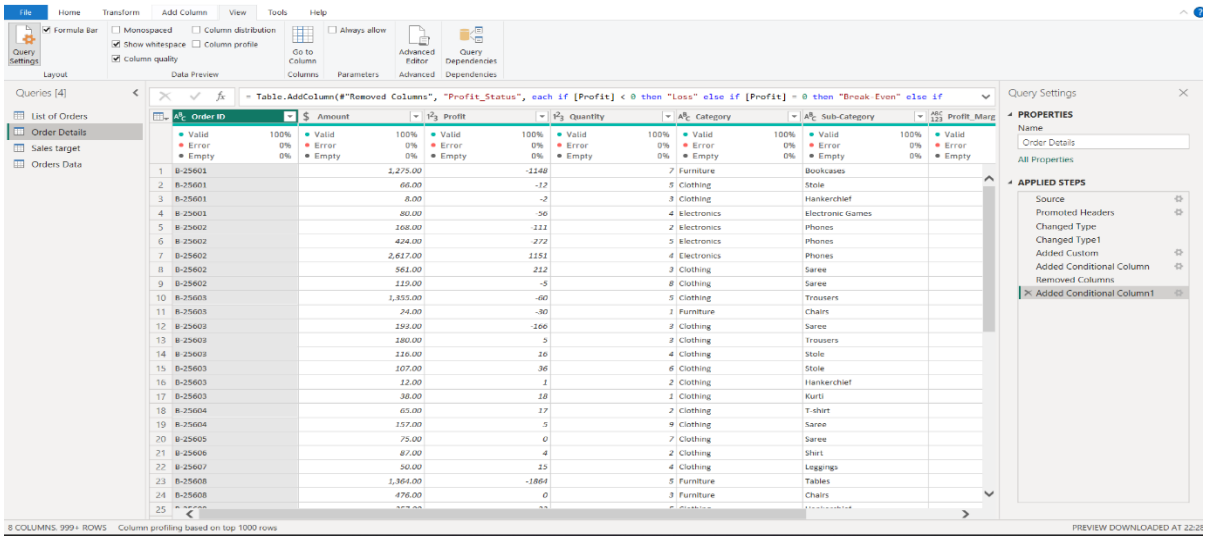
In the given dataset , the NULL values are present in the last 60 rows of the LIST OF ORDERS tsble. During the STEP-2 process , the LIST OF ORDERS table is restricted to top 500 rows so the NULL valued rows are removed.

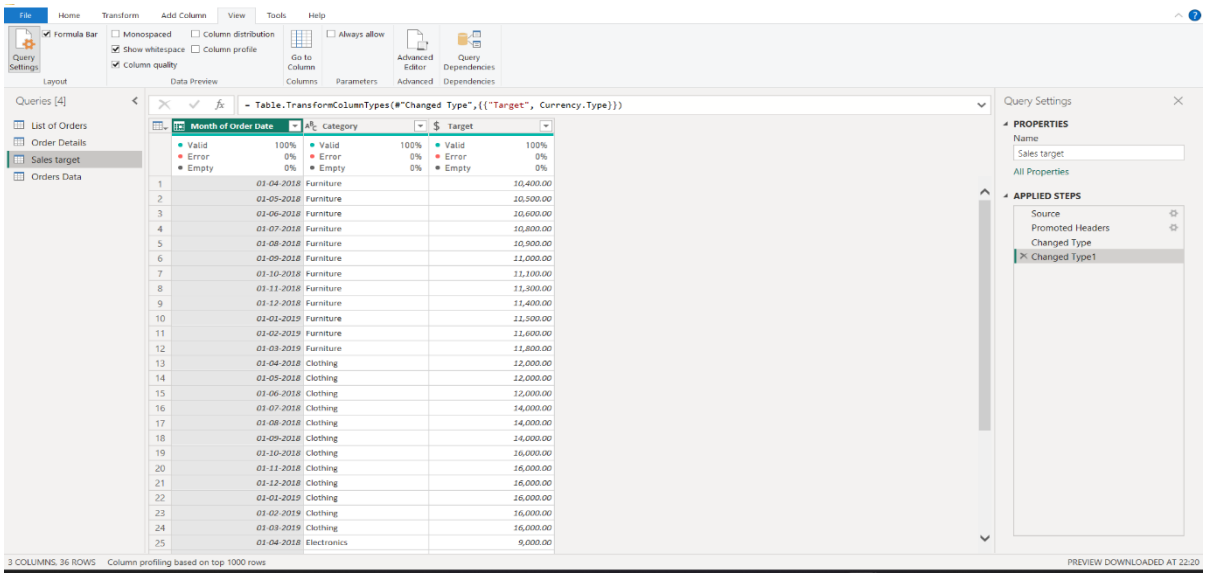
STRATEGY

By checking the quality of the each column in the tables , we can ehandle the missing data and duplicate data easily.

In the Power Query Editor , go to the VIEW tab, in the DATA PREVIEW group , select the COLUMN QUALITY checkbox.



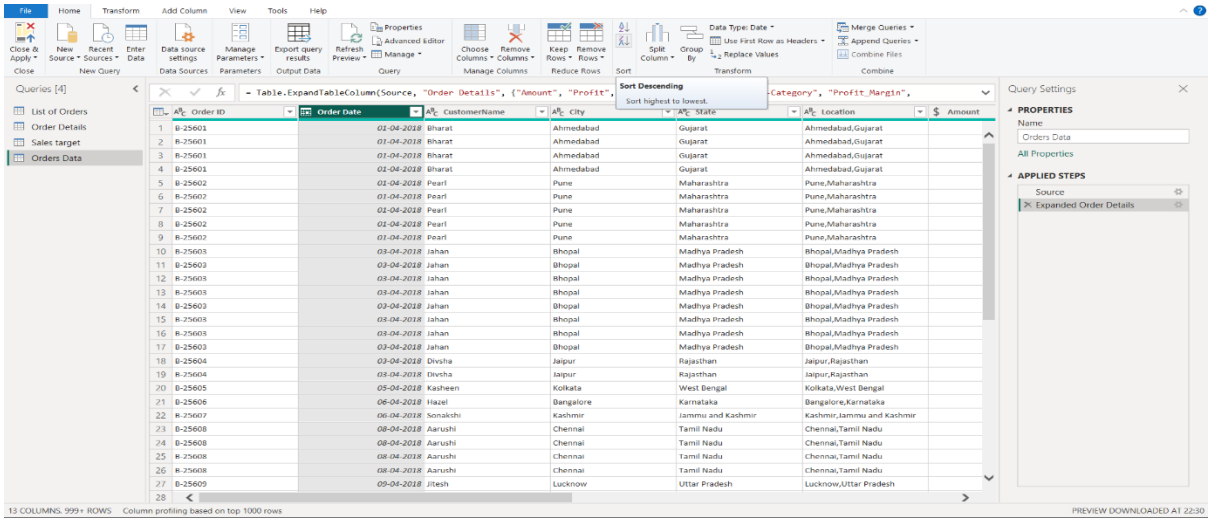


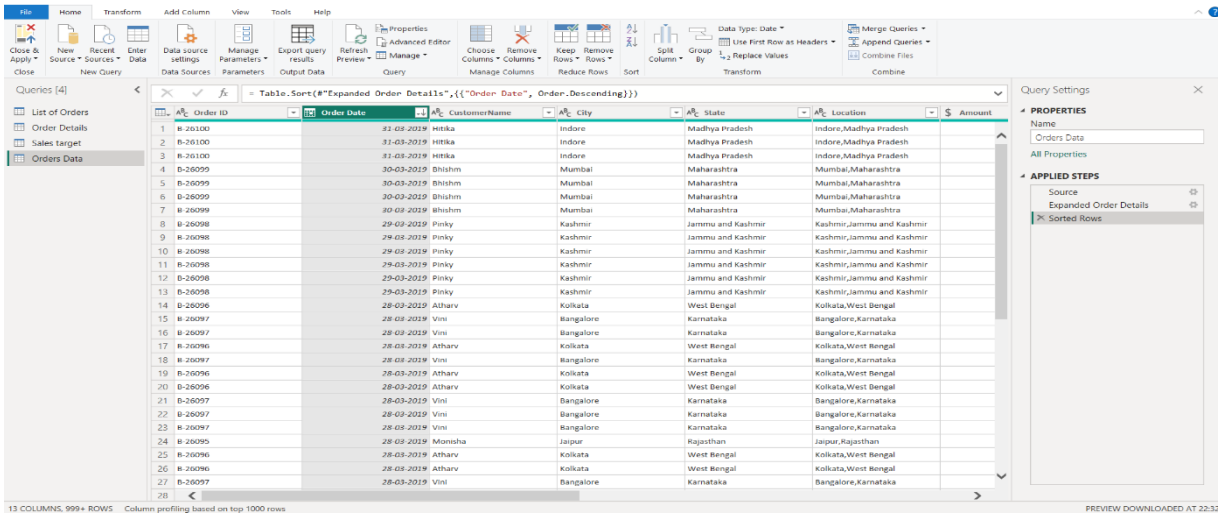


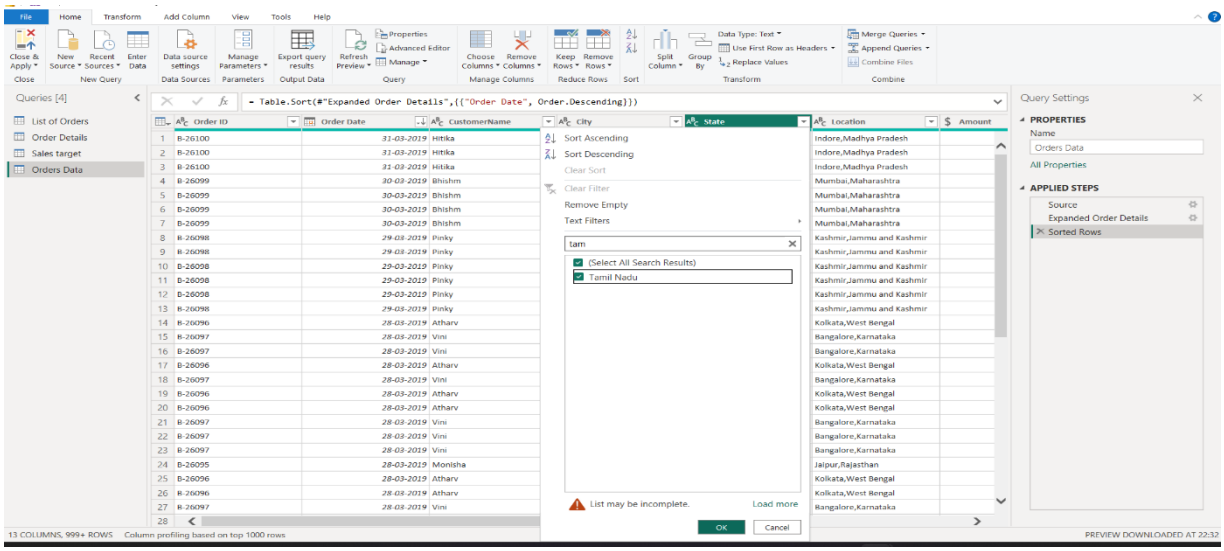
In the given dataset, no missing datas and duplicate datas are found.

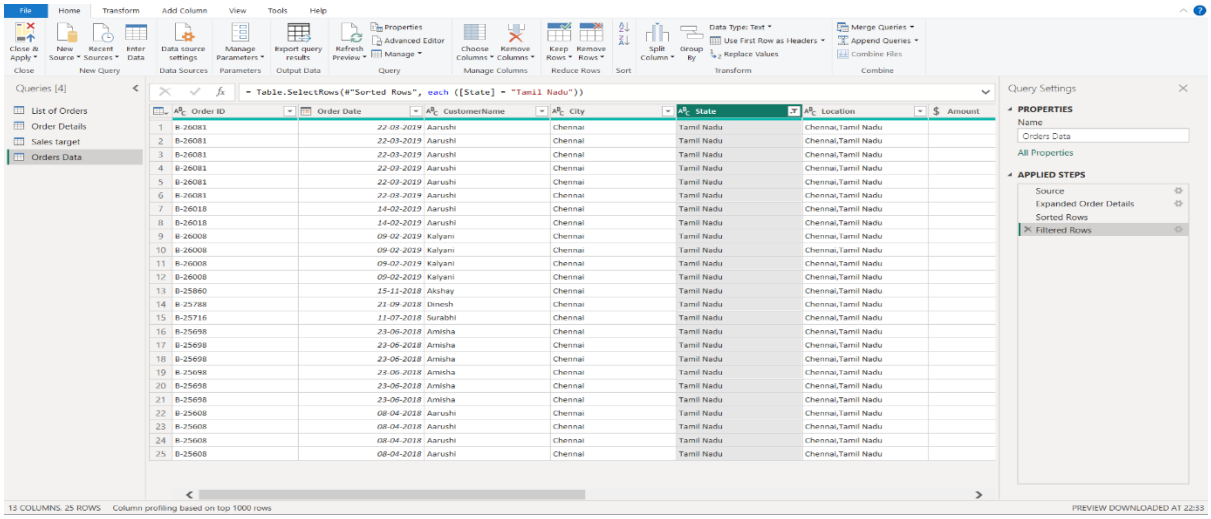
STEP-5----------SORTING AND FILTERING DATA----------

In the ORDERS DATA table, sorting techniques on column ORDER DATE is applied in the DESCENDING Manner to analyze the recent trends.
The orders are focused using Filter technique only on a specific state (e.g., Tamil Nadu) for regional analysis.


Sort Descending
Sort highest to lowest.
Queries [4]
List of Orders
Order Details
Sales target
Orders Data
Query Settings
PROPERTIES
Name
Orders Data
All Properties
APPLIED STEPS
Source
Expanded Order Details
13 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 22:30

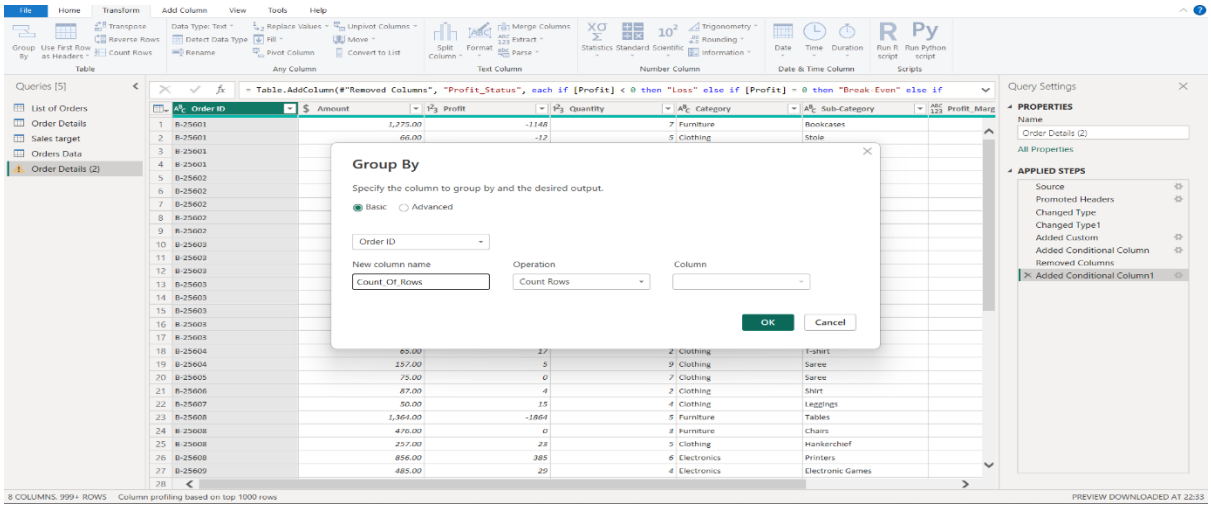

= Table.Sort(#"Expanded Order Details",{{"Order Date", Order.Descending}})
Queries [4]
List of Orders
Order Details
Sales target
Orders Data
Query Settings
PROPERTIES
Name
Orders Data
All Properties
APPLIED STEPS
Source
Expanded Order Details
Sorted Rows
13 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 22:32

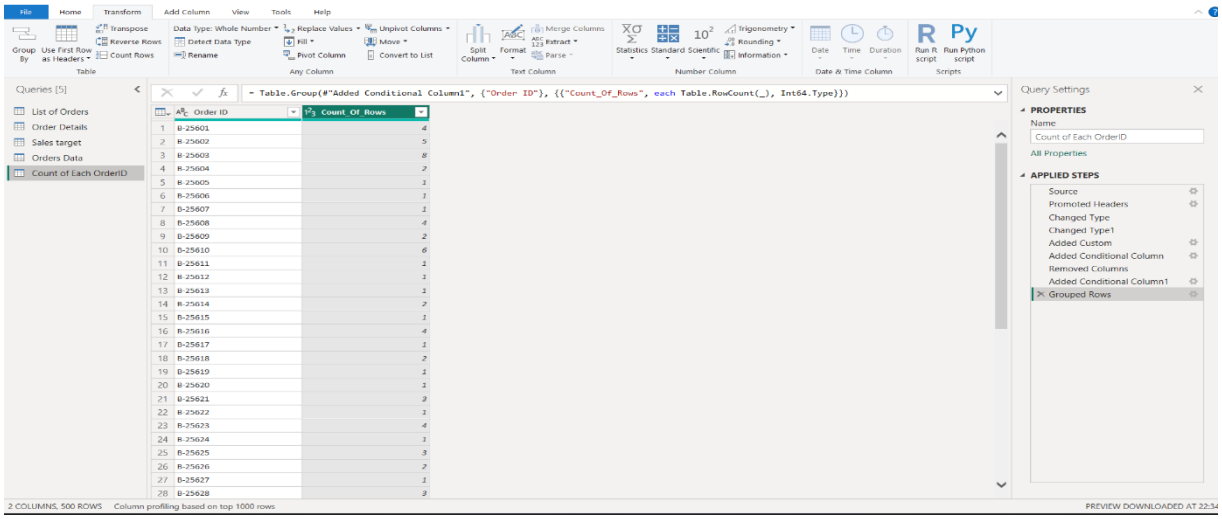

= Table.Sort(#"Expanded Order Details",{{"Order Date", Order.Descending}})
Sort Ascending
Sort Descending
Clear Sort
Clear Filter
Remove Empty
Text Filters
tam
(Select All Search Results)
Tamil Nadu
List may be incomplete.
Load more
OK
Cancel
Queries [4]
List of Orders
Order Details
Sales target
Orders Data
Query Settings
PROPERTIES
Name
Orders Data
All Properties
APPLIED STEPS
Source
Expanded Order Details
Sorted Rows
13 COLUMNS, 999+ ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 22:32

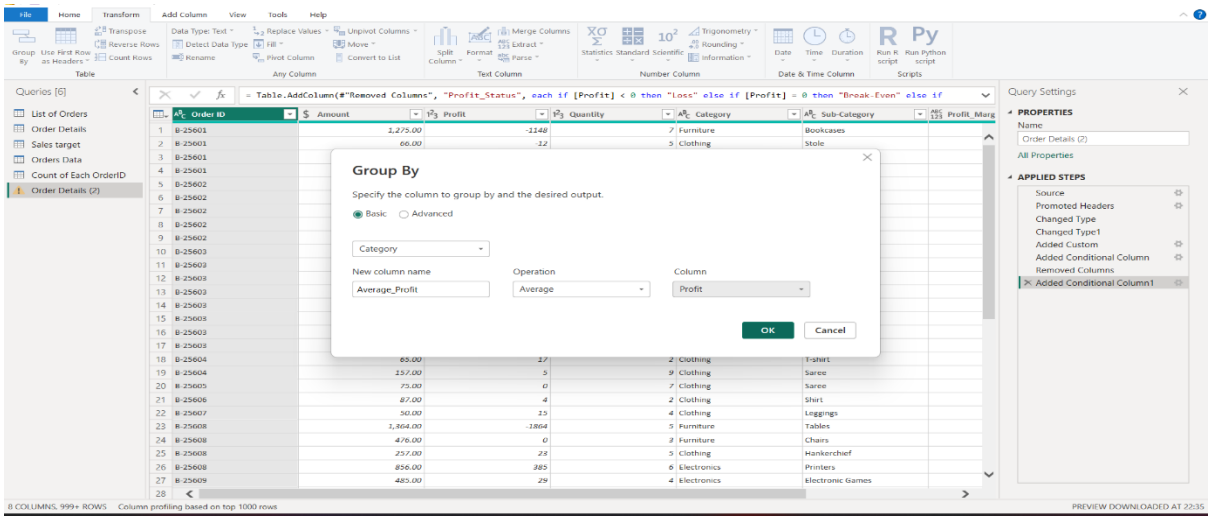


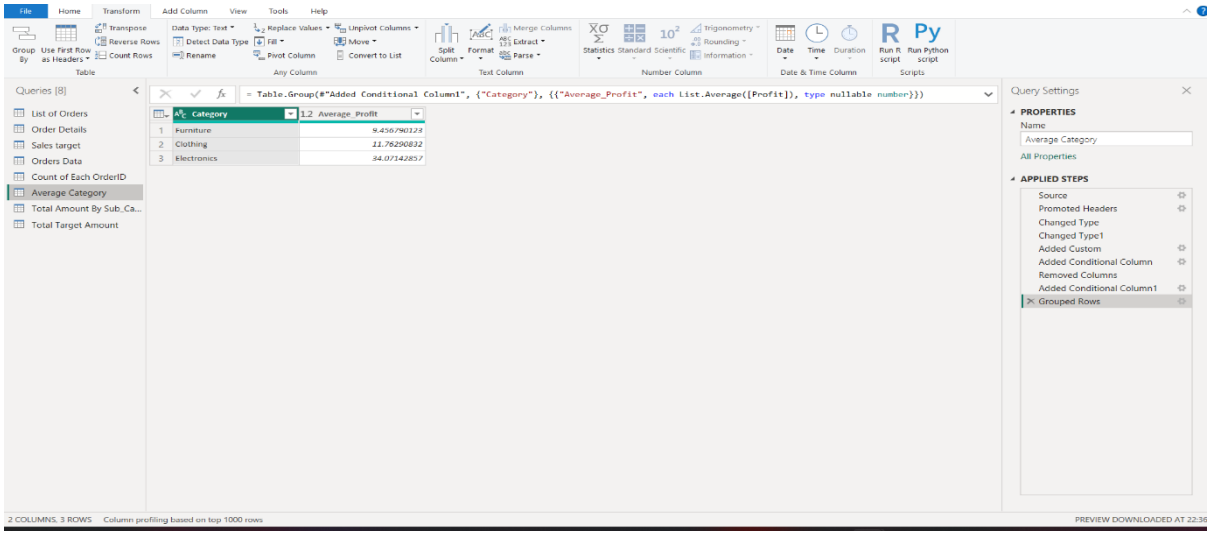
STEP-6----------GROUPING AND AGGREGATING DATA----------

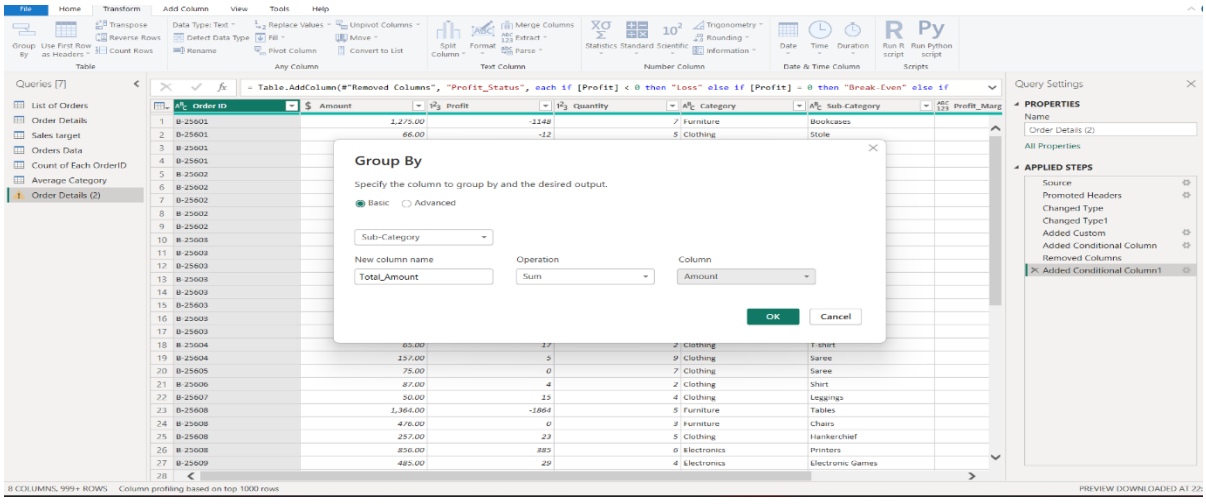
The ORDER DETAILS table is duplicated each time and THE COUNT OF EACH ORDER ID, AVERAGE PROFIT BY CATEGORY AND TOTAL AMOUNT BY SUB-CATEGORY is calculated using the GROUP BY function.

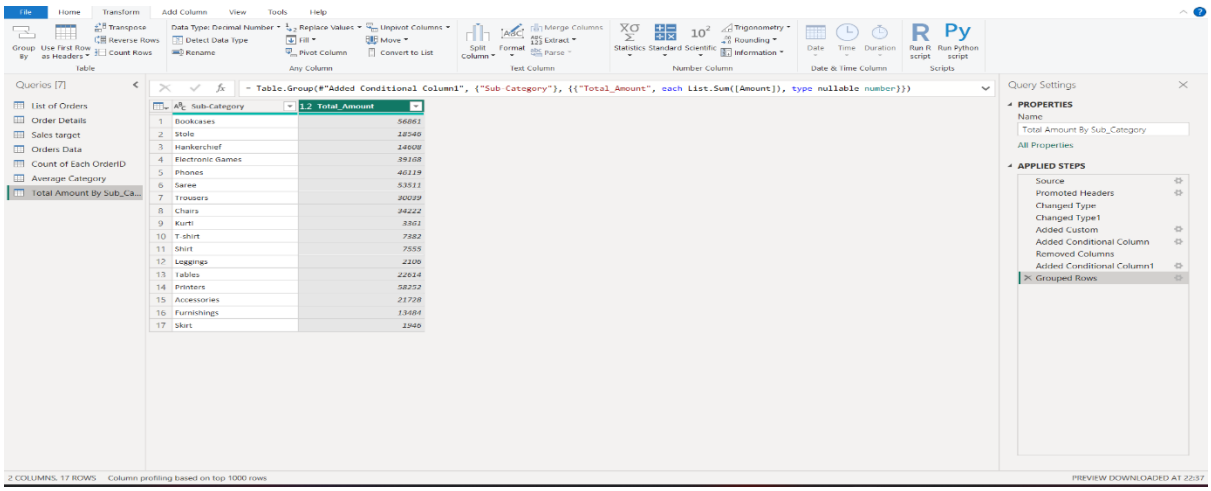




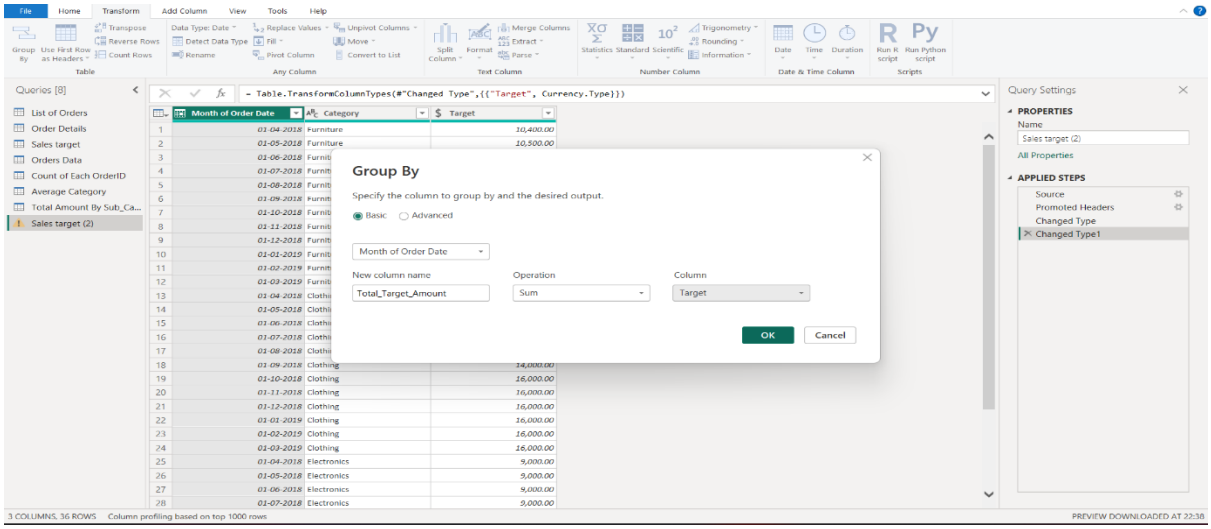

Group By
Specify the column to group by and the desired output.
Basic
Advanced
Category
New column name
Average_Profit
Operation
Average
Column
Profit
OK
Cancel

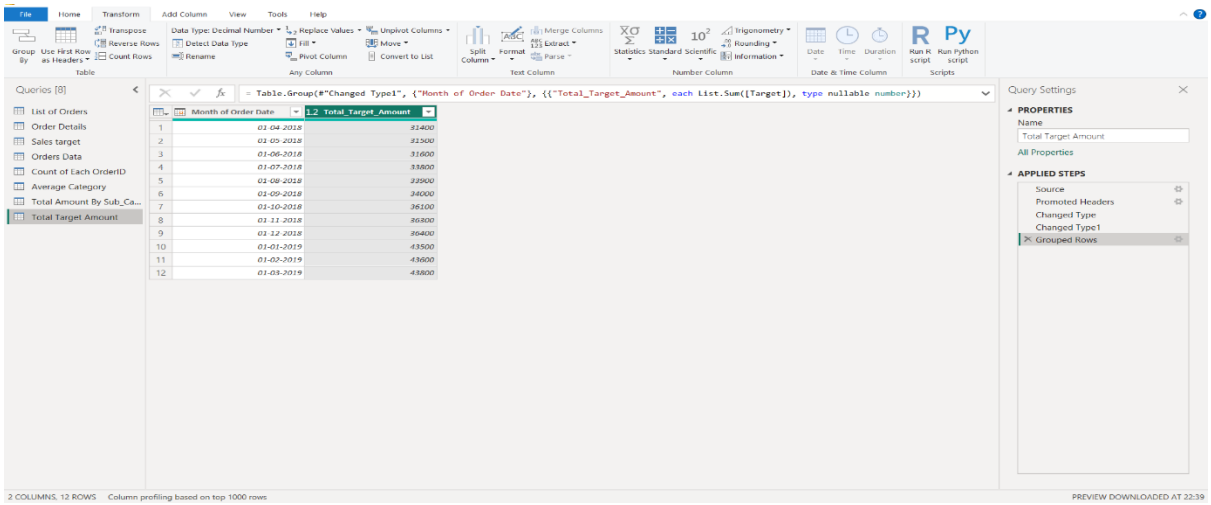



Group By
Specify the column to group by and the desired output.
Basic
Advanced
Sub-Category
New column name
Total_Amount
Operation
Sum
Column
Amount
OK
Cancel



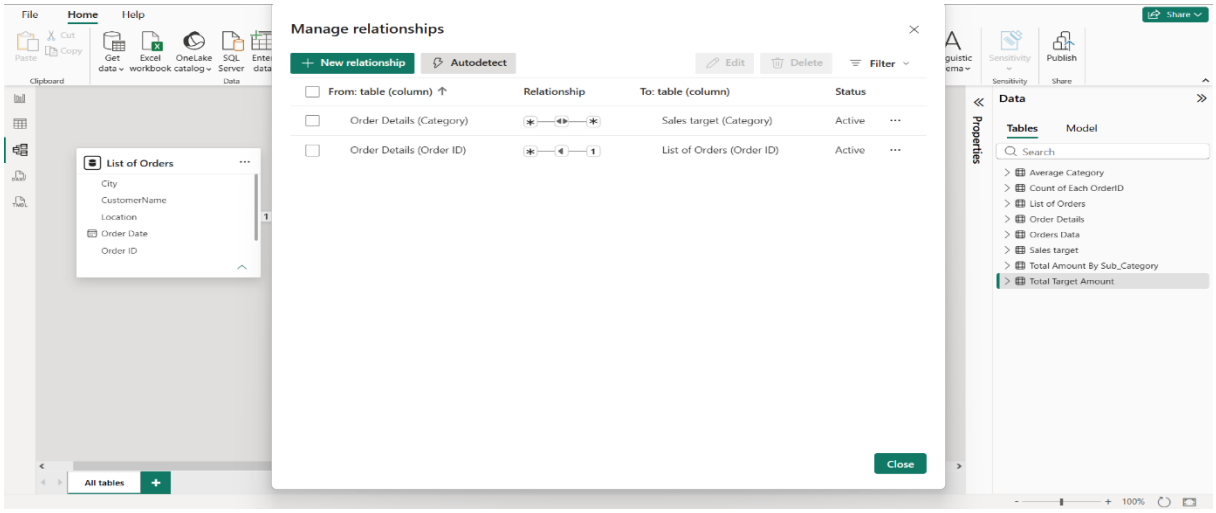
Now,the SALES TARGET table is duplicated and the TOTAL TARGET AMOUNT BY MONTH OF ORDER DATE is aggregated using the GROUP BY function.

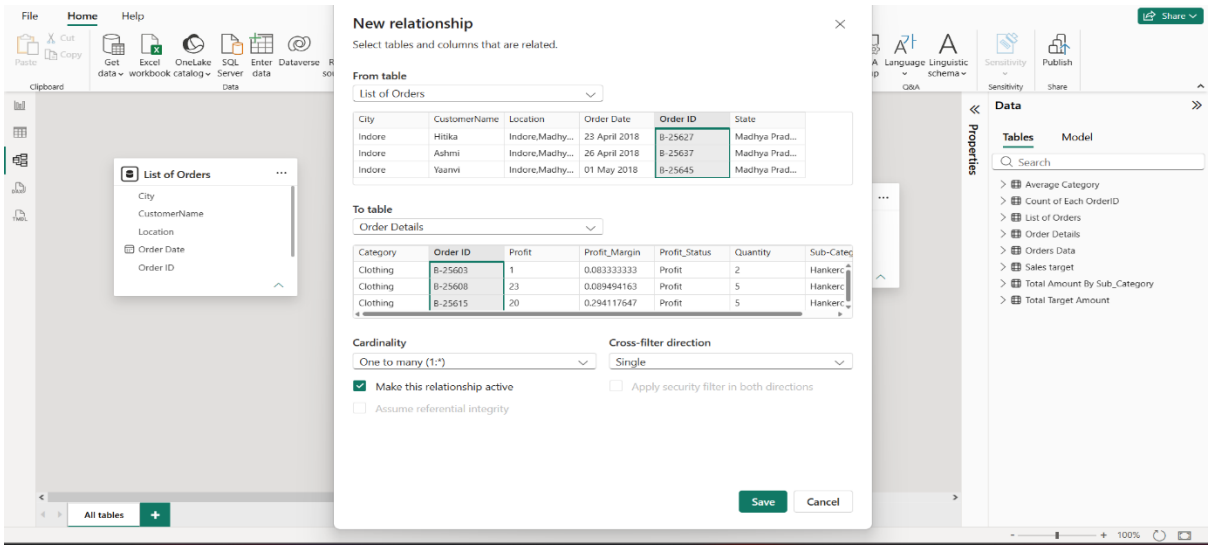


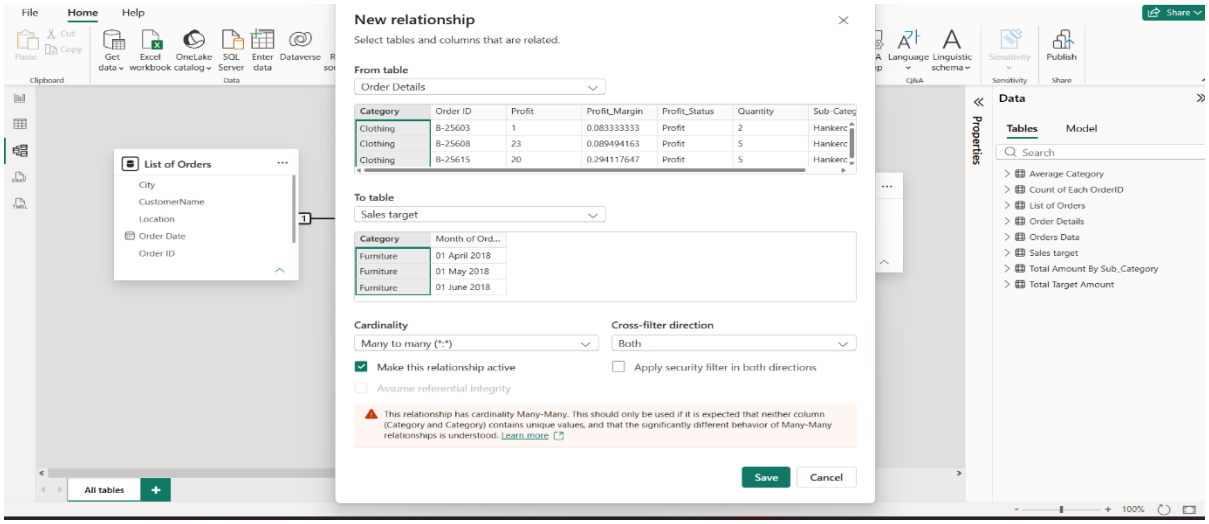


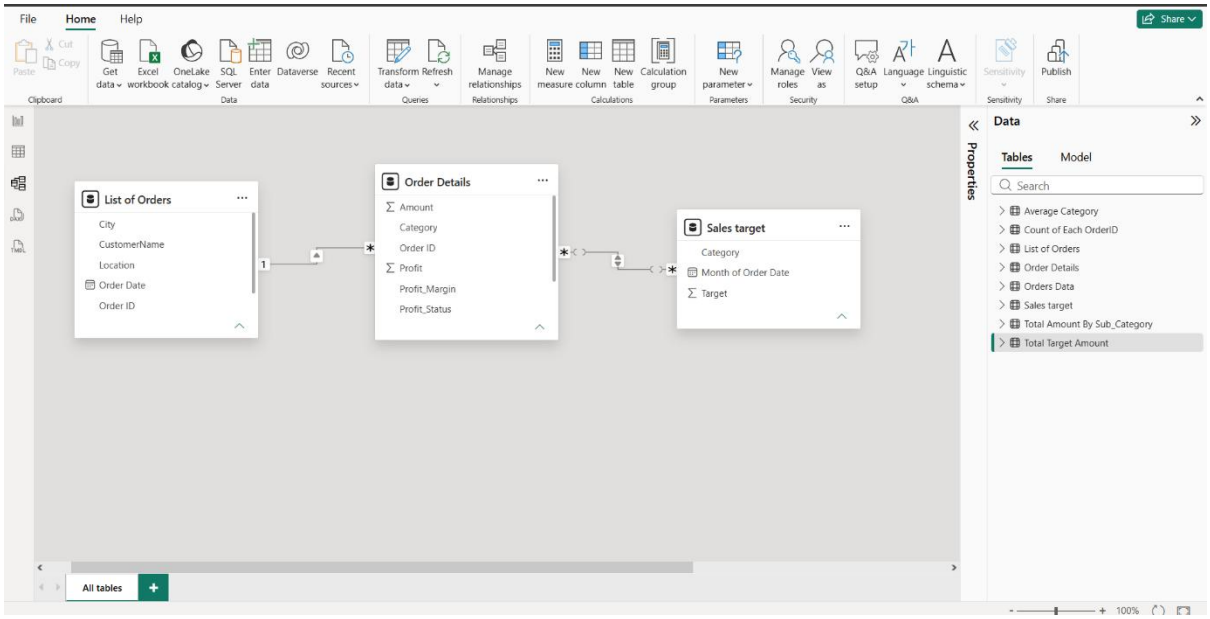
STEP-7----------DATA MODELING----------

The relationship between the LIST OF ORDERS and ORDER DETAILS tables are established using the ORDER ID column.
The relationship between the ORDER DETAILS and SALES TARGET tables are established using the CATEGORY column.
Click on Manage Relationships and the active relationships are ensured.






File
Home
Help
New relationship
Select tables and columns that are related.
From table
Order Details
Category
Order ID
Profit
Profit_Margin
Profit_Status
Quantity
Clothing
B-25603
1
0.083333333
Profit
2
B-25608
23
0.089494163
5
B-25615
20
0.294117647
To table
Sales target
Month of Ord...
Furniture
01 April 2018
01 May 2018
01 June 2018
Cardinality
Many to many (*:*)
Cross-filter direction
Both
Make this relationship active
Apply security filter in both directions
Assume referential integrity
This relationship has cardinality Many-Many. This should only be used if it is expected that neither column (Category and Category) contains unique values, and that the significantly different behavior of Many-Many relationships is understood. Learn more
Save
Cancel
List of Orders
City
CustomerName
Location
Order Date
Order ID
All tables
Data
Tables
Model
Search
Average Category
Count of Each OrderID
List of Orders
Order Details
Orders Data
Sales target
Total Amount By Sub_Category
Total Target Amount
Properties
100%


File
Home
Help
Share
Get data
Excel workbook
OneLake catalog
SQL Server
Enter data
Dataverse
Recent sources
Transform data
Refresh
Manage relationships
New measure
New column
New table
Calculation group
New parameter
Manage roles
View as
Q&A setup
Language
Linguistic schema
Sensitivity
Publish
List of Orders
City
CustomerName
Location
Order Date
Order ID
Order Details
Amount
Category
Order ID
Profit
Profit_Margin
Profit_Status
Sales target
Category
Month of Order Date
Target
All tables
Data
Tables
Model
Search
Average Category
Count of Each OrderID
List of Orders
Order Details
Orders Data
Sales target
Total Amount By Sub_Category
Total Target Amount
Properties
100%